

SutureFormer: Learning Surgical Trajectories via Goal-conditioned Offline RL in Pixel Space

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Abstract. Predicting surgical needle trajectories from endoscopic video is critical for robot-assisted suturing, enabling anticipatory planning, real-time guidance, and safer motion execution. Existing methods that directly learn motion distributions from visual observations tend to overlook the sequential dependency among adjacent motion steps. Moreover, sparse waypoint annotations often fail to provide sufficient supervision, further increasing the difficulty of supervised or imitation learning methods. To address these challenges, we formulate image-based needle trajectory prediction as a sequential decision-making problem, in which the needle tip is treated as an agent that moves step by step in pixel space. This formulation naturally captures the continuity of needle motion and enables the explicit modeling of physically plausible pixel-wise state transitions over time. From this perspective, we propose SutureFormer, a goal-conditioned offline reinforcement learning framework that leverages sparse annotations to dense reward signals via cubic spline interpolation, encouraging the policy to exploit limited expert guidance while exploring plausible future motion paths. SutureFormer encodes variable-length clips using an observation encoder to capture both local spatial cues and long-range temporal dynamics, and autoregressively predicts future waypoints through actions composed of discrete directions and continuous magnitudes. To enable stable offline policy optimization from expert demonstrations, we adopt Conservative Q-Learning (CQL) with Behavioral Cloning (BC) regularization. Experiments on a new kidney wound suturing dataset containing 1,158 trajectories from 50 patients show that SutureFormer reduces Average Displacement Error (ADE) by 58.6% compared with the strongest baseline, demonstrating the effectiveness of modeling needle trajectory prediction as pixel-level sequential action learning.

Keywords: Surgical Trajectory Prediction · Offline Reinforcement Learning · Robot-Assisted Surgery · Endoscopic Video Analysis

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1 Introduction

Robot-assisted surgery is gradually evolving from pure tele-operation toward task-level autonomy, where intelligent systems are expected to anticipate surgical intent and provide proactive assistance [25,1]. Within this paradigm, predicting instrument trajectories (e.g., needle), particularly during suturing, one of the most technically demanding and outcome-sensitive maneuvers in minimally invasive procedures. Accurate future trajectory estimation can support anticipatory planning, real-time guidance, and safer motion execution. From the perspective of bounded rationality [20], surgeons operate under inherent cognitive constraints including limited attention span, finite working memory, and time pressure collectively bound the optimality of intraoperative decisions. A learning-based trajectory prediction system that distills expert demonstrations into anticipatory guidance can therefore serve as a decision-support mechanism, helping less experienced surgeons approximate expert-level performance and ultimately improving access to high-quality surgical care.

Despite growing interest in deep learning for surgical scene analysis [14], including workflow recognition [10] and scene understanding [15], research on precise procedural assistance for trajectory prediction remains nascent. Currently, most approaches depend on robot kinematic signals, including joint angles, end-effector poses, and gripper states [17,19,23]. Such data are available only on platforms with accessible kinematic interfaces (e.g., the da Vinci Research Kit) and often require direct collaboration with device manufacturers. This dependency fundamentally limits transferability: a policy trained on one robotic platform cannot generalize to another, let alone to the vast archive of conventional laparoscopic video where no kinematic readout exists. In addition, many methods require dense temporal annotations, which are prohibitively expensive to obtain.

Learning control directly from vision is a challenging but transformative idea, one that has seen remarkable success in adjacent fields. For instance, purely vision-based imitation learning agents have mastered complex control tasks like driving, learning to navigate and plan by directly mapping image inputs to steering commands [16,3]. Cai et al. [4] demonstrated that cost functions for reinforcement learning (RL) can be learned directly from images, bypassing the need for explicit state estimation. Similarly, Tamar et al. proposed Value Iteration Networks (VIN) [21] that a neural network can learn to perform goal-directed reasoning, generalizing its policy to novel environments by embedding a computational process akin to planning within its architecture. These insights suggest that the "kinematic bottleneck" in surgical trajectory prediction is not a technological inevitability, but rather a design choice, although it is more challenging in fine-grained surgical manipulation scenarios.

Recent work on surgical trajectory prediction has only just emerged, indicating that such high-level assistance tasks can support preoperative training and intraoperative guidance but still remain underexplored. For example, Li et al. [12] introduced imitation learning for dissection trajectory prediction, using an implicit diffusion policy (iDiff-IL) to model a joint state-action distribution to capture uncertainty in future dissection trajectories; but still face challenges in

predicting uncertain movements and generalizing to various scenes. Subsequent work has attempted to learn motion distributions directly from surgical videos. Zhao et al.’s multi-scale phase-conditioned diffusion (MS-PCD) [26] cascades diffusion models across multiple scales to generate conditioned action sequences for procedure planning. Xu et al. proposed the DP4AuSu, applying diffusion policies to autonomous suturing with dynamic time warping and locally weighted regression to capture multimodal demonstration trajectories, achieving 94% insertion success in simulation and 85% suture success on a real robot [24]. However, these diffusion-based generative approaches either predict an entire trajectory in one pass or depend on slow autoregressive decoding, which increases inference complexity and can accumulate errors. More importantly, they fail to explicitly model stepwise motion dependency, and their reliance on sparse annotations provides insufficient supervision for long-horizon trajectory learning.

These limitations motivate us a different modeling approach: rather than casting image-driven needle trajectory prediction as a one-off regression or generative problem, we formulate it as a stepwise decision-making process in which the needle tip is treated as an agent progressing through pixel space. This view better conforms to the physical continuity of motion and allows explicit modeling of pixel-wise state transitions; sparse annotations can be exploited as reward signals rather than direct supervision, making Reinforcement Learning (RL) a natural choice for learning from limited guidance. RL has already been explored in surgical robotics: Ji et al. [9] proposed a heuristically accelerated deep Q network for safe planning of neurosurgical flexible needle insertion paths, demonstrating that RL can plan complex trajectories but requires many training episodes; Lin et al. [13] built a world-model-driven pixel-level deep RL framework (GAS) that achieves robust grasping for various surgical instruments. Nevertheless, RL has not been fully exploited for image-based needle trajectory prediction. In this work, we leverage RL to model the dependency between adjacent motion steps and to learn effective trajectory policies from sparse supervision.

To this end, we propose SutureFormer, which reformulates surgical trajectory prediction as goal-conditioned visual navigation in pixel space using offline expert demonstrations. A goal-conditioned policy iteratively predicts future needle waypoints based solely on local visual context and sparse keyframe guidance. The observation encoder combines a Spatial CNN and Transformer to capture spatial cues and long-range temporal dependencies. A discrete-action Conservative Q-Learning (CQL) agent then autoregressively generates trajectory waypoints using a 9 direction action space with continuous step magnitudes, conditioned on keyframe goals during training and polynomial extrapolation at inference. In practice, SutureFormer requires only endoscopic video frames and 9 sparse keyframes annotations per trajectory. To our knowledge, this is the first framework to formulate surgical trajectory prediction as visual navigation task with offline RL in pixel space. Our method outperforms than current state-of-art methods, demonstrating the viability of offline RL for surgical trajectory prediction.

In summary, our contributions are threefold:

- We reformulate image-based surgical needle trajectory prediction as a sequential decision-making problem in pixel space, providing a new perspective that explicitly captures motion continuity and stepwise state transitions.
- We propose a goal-conditioned offline RL framework that converts sparse waypoint annotations into dense reward signals via cubic spline interpolation, allowing limited supervision to guide policy learning more effectively than direct supervised or imitation-based formulations.
- We validate the proposed method on a new kidney wound suturing dataset with 1,158 trajectories from 50 patients, where it achieves substantial improvements over strong baselines, demonstrating the promise of our paradigm for surgical trajectory prediction.

2 Methods

2.1 Problem Statement and Annotation Interpolation

Given a variable-length observed video segment $V_{0:T_{\text{obs}}} = \{I_t\}_{t=0}^{T_{\text{obs}}}$ and the corresponding needle tip positions $\{(x_t, y_t)\}_{t=0}^{T_{\text{obs}}}$, the goal is to predict the future trajectory $\{(\hat{x}_t, \hat{y}_t)\}_{t=T_{\text{obs}}+1}^{T_{\text{obs}}+T_{\text{pred}}}$. We cast this task as goal-conditioned sequential prediction in pixel space and learn the predictor entirely from an offline dataset of expert demonstrations. Specifically, each training trajectory is transformed into a sequence of state-transition tuples extracted fully from annotated expert motion without any online interaction. The overall framework consists of an observation encoder that summarizes visual-temporal context and a goal-conditioned policy head that autoregressively predicts future waypoint displacements.

To obtain dense per-frame annotations, we independently apply cubic spline interpolation [5] to the x and y coordinate sequences as functions of the frame index. Given the 9 keyframe positions, we fit two natural cubic spline functions, $S_x(t)$ and $S_y(t)$, evaluating them at every intermediate frame within the temporal range. The interpolated coordinates are rounded to the nearest integer to yield valid pixel locations, without extrapolation beyond the first and last keyframes. Each interpolated frame receives a confidence score from 0.45 to 0.9 based on its temporal proximity to the nearest keyframe, assigning higher scores to closer frames. The expert annotated keyframes retain a confidence score of 1.0.

2.2 Observation Encoder

At each observed time step, we extract a 128×128 RGB crop centered at the annotated needle-tip location in the full image. To complement local appearance cues with sparse trajectory supervision, inspired by G2RL [22], we construct an additional guidance channel over the same crop by rasterizing the available trajectory path into a single-channel heatmap. The heatmap intensity is modulated by annotation confidence, with manually annotated keyframes assigned higher confidence than interpolated intermediate positions. The RGB crop and

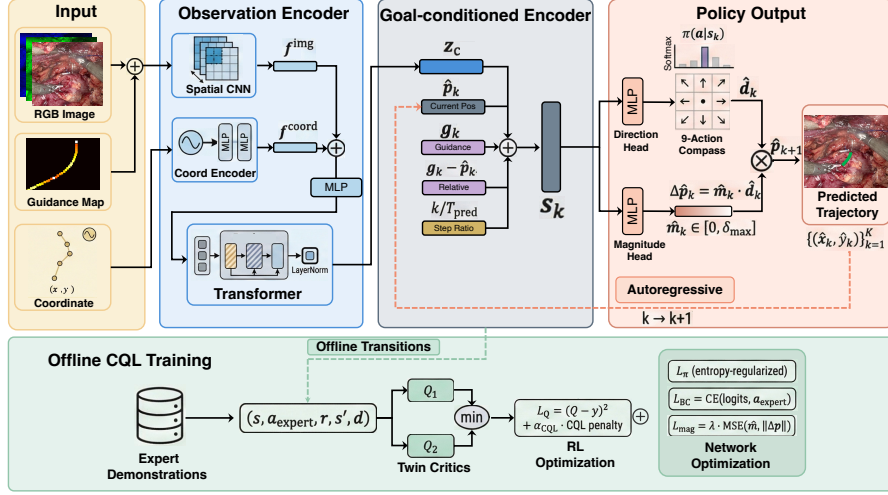


Fig. 1. Overview of the proposed framework. (i) Given the observed video segment, the observation encoder extracts local visual guidance features from needle-centered crops and aggregates their temporal dependencies with a Transformer to obtain the contextual representation z_c . (ii) At each prediction step k , the goal-conditioned state encoder constructs the state s_k by combining z_c with the encoded current position $\hat{p}_k$, guidance coordinate g_k , relative displacement $g_k - \hat{p}_k$, and normalized step ratio k/T_{pred} . (iii) The policy head then predicts a discrete motion direction and a continuous step magnitude to autoregressively update the needle-tip position and generate the future trajectory. (iv) Training is performed entirely offline on expert transitions using Conservative Q-Learning with twin critics, together with auxiliary behavior-cloning and magnitude-regression objectives.

the guidance map are concatenated to form a 4-channel input for each observed frame.

All 4-channel crops are concatenated and processed by a spatial feature extractor to obtain a unified embedding, which is then fed into a Transformer encoder to model temporal dependencies across the observation window. To support variable-length observation sequences, masked attention is applied so that the encoder attends only to valid time steps and ignores padded positions. In addition, we adopt bucketed batch sampling to reduce excessive padding and improve training efficiency. The final contextual representation, denoted by z_c , summarizes the visual and temporal evidence available up to the current prediction step.

2.3 Goal-conditioned Encoder

At prediction step k , the policy operates on a state vector

$$s_k = \phi(z_c, \hat{p}_k, g_k, g_k - \hat{p}_k, k/T_{pred}) \quad (1)$$

where z_c is the contextual representation produced by the observation encoder, $\hat{p}_k$ is the current predicted position, g_k is a step-specific guidance coordinate, $g_k - \hat{p}_k$ is the relative displacement from the current position to the guidance target, and k/T_{pred} encodes the normalized prediction progress. Three independent sinusoidal coordinate encoders are used to embed $\hat{p}_k$, g_k , and $g_k - \hat{p}_k$, respectively, while a linear layer projects the scalar progress ratio into the same feature space. Their concatenation with z_c is then mapped by $\phi(\cdot)$ to the final state representation s_k .

We parameterize each motion step using a discrete direction and a continuous magnitude. The discrete action space contains nine actions: eight movement directions uniformly spaced at 45° intervals and one idle action. Let $a \in \{1, \dots, 9\}$ denote the direction action. The corresponding unit vectors are defined as:

$$u_a = \begin{cases} \left[\sin\left(\frac{(a-1)\pi}{4}\right), -\cos\left(\frac{(a-1)\pi}{4}\right) \right]^\top, & a = 1, \dots, 8 \\ (0, 0)^\top, & a = 9 \end{cases} \quad (2)$$

The policy predicts a categorical distribution $\pi(a \mid s_k)$ over the nine direction actions, while a separate magnitude head outputs a non-negative step length $\hat{m}_k \in [0, \delta_{\max}]$. During inference, we compute the expected direction vector:

$$\hat{d}_k = \sum_{a=1}^9 \pi(a \mid s_k) u_a \quad (3)$$

and generate the next displacement and position as:

$$\Delta \hat{p}_k = \hat{m}_k \hat{d}_k, \quad \hat{p}_{k+1} = \text{clip}(\hat{p}_k + \Delta \hat{p}_k, 0, 1) \quad (4)$$

where clipping enforces the normalized image-coordinate constraint $[0, 1]^2$. This formulation yields smooth autoregressive motion while retaining a discrete directional action space for value learning.

The guidance coordinate g_k is defined differently for training and inference. During training, g_k is taken from the ground-truth future trajectory available in the expert demonstration. During inference, future ground-truth positions are unavailable; therefore, we replace them with pseudo-guidance obtained by polynomial extrapolation from the last L observed points ($L = 10$ by default). A quadratic least-squares fit is used when at least five observed points are available; otherwise, linear extrapolation is used. The extrapolated coordinates are also clipped to $[0, 1]^2$. Stated differently, the model is trained with oracle future guidance and evaluated with an extrapolated test-time surrogate.

2.4 Offline Conservative Q-Learning

The model is trained entirely offline from a static expert trajectories. For each training trajectory, we traverse the annotated sequence and extract expert transitions of the form $(s_k, a_k, r_k, s_{k+1}, d_k)$, where a_k is obtained by quantizing the

expert displacement to its nearest compass direction, r_k is the step reward, and d_k indicates episode termination. The next state s_{k+1} is constructed from the successor position in the demonstration trajectory rather than generated through online environment interaction. This setup corresponds to offline reinforcement learning on logged expert transitions.

To learn conservative action values from fixed offline data, we adopt twin Q-functions optimized with Conservative Q-Learning (CQL) [11]. The critic objective combines a Bellman regression term with a conservative regularizer that suppresses Q-values for unsupported actions:

$$\mathcal{L}_{Q_i} = \mathbb{E}_{(s,a,r,s',d) \sim \mathcal{D}} \left[(Q_i(s,a) - y)^2 \right] + \alpha_{\text{cql}} \mathbb{E}_{s \sim \mathcal{D}} \left[\log \sum_a \exp(Q_i(s,a)) - Q_i(s, a_{\text{exp}}) \right] \quad (5)$$

Here, a_{exp} denotes the expert action and α_{cql} controls the strength of the conservative penalty. The soft value target is computed as

$$V(s') = \sum_{a'} \pi(a'|s') \left[\min(Q_1^{\text{tgt}}(s', a'), Q_2^{\text{tgt}}(s', a')) - \alpha \log \pi(a'|s') \right] \quad (6)$$

with entropy temperature $\alpha = 0.2$. The policy is optimized using the entropy-regularized objective

$$\mathcal{L}_{\pi} = \mathbb{E}_{s \sim \mathcal{D}} \left[\sum_a \pi(a|s) (\alpha \log \pi(a|s) - \min(Q_1(s,a), Q_2(s,a))) \right] \quad (7)$$

Following our implementation, policy learning is further stabilized by a behavior-cloning term

$$\mathcal{L}_{\text{BC}} = \text{CE}(\text{logits}, a_{\text{exp}}) \quad (8)$$

and the magnitude head is supervised with

$$\mathcal{L}_{\text{mag}} = \lambda_{\text{mag}} \mathbb{E} \left[(\|\Delta \hat{p}_k\|_2 - \|p_{k+1} - p_k\|_2)^2 \right]. \quad (9)$$

Thus, the final training objective combines conservative value learning, expert-action imitation, and direct step-length supervision.

A key implementation detail is gradient-flow separation. Specifically, the critic consumes state features with detached gradients so that critic updates do not propagate into the observation encoder. In contrast, gradients from the actor loss and the magnitude-regression loss are allowed to update the shared encoder. This choice is motivated by the offline setting: the CQL critic is explicitly regularized to be pessimistic on unsupported actions, allowing critic gradients to dominate the encoder may bias the representation toward conservative value suppression rather than accurate motion modeling. We therefore restrict encoder updates to objectives that are directly tied to displacement prediction.

2.5 Reward Design under Sparse Keyframe Supervision

With only nine keyframes are manually annotated for each trajectory, we densify supervision by fitting a cubic spline to the annotated 2D coordinates and evaluating it at intermediate frames. This produces dense frame-wise reference positions

$\{\tilde{p}_k\}_{k=0}^{K-1}$ for reward construction, while preserving the original keyframes as the most reliable supervisory anchors. Each interpolated frame is further assigned a confidence score according to its temporal proximity to the nearest manually annotated keyframe; manually annotated keyframes are assigned confidence 1.0, whereas interpolated positions receive lower confidence values.

At prediction step k , let $\hat{p}_k$ denote the predicted position and $\tilde{p}_k$ the corresponding dense reference position. Their Euclidean distance in normalized image coordinates is defined as

$$d_k = \|\hat{p}_k - \tilde{p}_k\|_2 \quad (10)$$

The reward at step k consists of three components: a constant time penalty, a confidence-weighted proximity term, and a terminal shaping term applied only at the final prediction step:

$$r_k = r_{\text{time}} + r_{\text{prox},k} + r_{\text{term},k} \quad (11)$$

where the time penalty is fixed as $r_{\text{time}} = -0.01$ and the proximity reward is defined as:

$$r_{\text{prox},k} = w_k r_{\text{prox},\max} \left(1 - \frac{d_k}{\tau}\right) \quad (12)$$

where $\tau = 0.02$ is the distance threshold in normalized coordinates and $r_{\text{prox},\max} = 0.5$.

To reflect annotation reliability, the confidence weight w_k is defined as

$$w_k = \begin{cases} 1.0, & \text{for manually annotated keyframes} \\ 0.5 + 0.5 \cdot \text{confidence}_k, & \text{for interpolated frames} \end{cases} \quad (13)$$

where $\text{confidence}_k \in [0, 1)$ denotes the confidence assigned to the interpolated reference position at step k .

To emphasize endpoint accuracy, we further introduce a terminal shaping term at the final prediction step:

$$r_{\text{term},k} = \begin{cases} w_k r_{\text{prox},\max} \left(1 - \frac{d_k}{\tau}\right), & k = K - 1 \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

where K is the total number of prediction steps for the current sample. This term shares the same distance-dependent form as the proximity reward, but is applied only at the final step so as to place additional emphasis on the accuracy of the predicted endpoint.

Overall, this reward design provides dense step-wise learning signals from sparsely annotated trajectories while down-weighting less reliable interpolated supervision.

3 Experiments

Datasets. We evaluate SutureFormer on clinical dataset of robotic-assisted laparoscopic kidney wound suturing operated by expert surgeon. The dataset comprises surgical videos from 50 patients and contains a total of 1,158 trajectories,

where clinical experts annotate 9 keyframes within each trajectory. The dataset is partitioned at the patient level into training, validation, and test sets, consisting of 35, 8, and 7 patients (corresponding to 861, 151, and 146 trajectories), respectively.

Implementation Details. The model is trained for 100 epochs with batch size 8 on an NVIDIA RTX 5090 GPU. We use four Adam optimizers with cosine annealing decaying to 1% of the initial learning rates: 1×10^{-4} for the observation encoder and 3×10^{-4} for the actor, critic, and magnitude heads. CQL hyperparameters are $\alpha_{\text{CQL}} = 0.01$, $\gamma = 0.95$, with soft target updates $\tau = 0.005$. Policy and magnitude updates subsample up to 2,048 transitions per batch.

Evaluation Metrics. All metrics are computed in pixel space by rescaling normalized coordinates to the original 1264×902 resolution. We report three complementary metrics (lower is better for all):

Average Displacement Error (ADE) measures the mean point-wise accuracy over the entire predicted trajectory:

$$\text{ADE} = \frac{1}{T_{\text{pred}}} \sum_{k=1}^{T_{\text{pred}}} \|\hat{p}_k - p_k\|_2 \quad (15)$$

Final Displacement Error (FDE) captures the positional accuracy at the trajectory endpoint, which is particularly relevant for suturing tasks where the needle must arrive at a precise target:

$$\text{FDE} = \|\hat{p}_{T_{\text{pred}}} - p_{T_{\text{pred}}}\|_2 \quad (16)$$

Discrete Fréchet Distance (FD) evaluates global shape similarity between the predicted and ground-truth trajectories, defined recursively as:

$$\begin{aligned} \text{dp}[i, j] &= \max \left(\min(\text{dp}[i-1, j], \text{dp}[i-1, j-1], \text{dp}[i, j-1]), \|\hat{p}_i - p_j\|_2 \right) \\ \text{FD} &= \text{dp}[T_{\text{pred}}, T_{\text{pred}}] \end{aligned} \quad (17)$$

All three metrics are averaged over the test set.

4 Results

4.1 Comparison with Baselines

To evaluate the effectiveness of our SutureFormer we compare against following baselines: (1) Behavioral Cloning (BC) [2], which encodes stacked frames via a U-Net [18] and regresses future coordinates with an MLP; (2) Generative Adversarial Imitation Learning (GAIL) [8], which generates trajectories conditioned on a random latent vector with a discriminator distinguishing real from generated pairs; (3) Implicit Behavioral Cloning (IBC) [6], an energy based model trained with InfoNCE loss and sampled via Langevin MCMC at inference; (4) Implicit

Table 1. Quantitative comparison on the test set under two settings (Obs = 6, Pred = 3 and Obs = 3, Pred = 6). Specifically, Obs = 6 denotes an observed surgical trajectory consisting of 6 annotated keyframes, with the remaining labels generated by interpolation at a frame interval of 1, whereas Pred = 3 denotes the prediction horizon containing 3 future keyframes. All metrics are evaluated in pixel space (lower is better), with the best results highlighted in **bold** and second best in underline.

Method	Seed	Obs = 6, Pred = 3			Obs = 3, Pred = 6		
		ADE ($\downarrow$)	FDE ($\downarrow$)	FD ($\downarrow$)	ADE ($\downarrow$)	FDE ($\downarrow$)	FD ($\downarrow$)
VanillaCNN	–	208.79 $\pm$ 93.65	221.39 $\pm$ 101.68	234.47 $\pm$ 100.29	194.86 $\pm$ 84.86	227.73 $\pm$ 112.16	249.74 $\pm$ 101.67
BC [2]	–	<u>128.15 $\pm$ 85.30</u>	<u>146.24 $\pm$ 100.87</u>	<u>156.83 $\pm$ 99.81</u>	<u>135.65 $\pm$ 75.23</u>	<u>182.31 $\pm$ 109.62</u>	<u>193.86 $\pm$ 104.57</u>
GAIL [8]	–	268.31 $\pm$ 128.76	281.42 $\pm$ 136.55	301.98 $\pm$ 133.52	242.24 $\pm$ 117.38	268.84 $\pm$ 141.19	314.21 $\pm$ 126.19
IBC [6]	–	233.84 $\pm$ 132.22	250.76 $\pm$ 150.40	276.25 $\pm$ 141.66	214.63 $\pm$ 105.23	250.88 $\pm$ 153.55	288.38 $\pm$ 136.58
iDiff-IL [12]	–	179.65 $\pm$ 104.41	197.80 $\pm$ 115.62	208.52 $\pm$ 114.79	185.78 $\pm$ 95.33	228.86 $\pm$ 134.08	246.57 $\pm$ 123.23
CondDiff [12]	–	155.02 $\pm$ 90.48	176.03 $\pm$ 105.68	186.28 $\pm$ 104.52	158.20 $\pm$ 90.33	203.38 $\pm$ 128.04	219.98 $\pm$ 119.80
MID [7]	–	148.64 $\pm$ 93.01	165.28 $\pm$ 107.93	185.39 $\pm$ 105.36	154.65 $\pm$ 80.46	213.77 $\pm$ 128.01	225.18 $\pm$ 120.95
Ours	42	56.01 $\pm$ 32.81	85.51 $\pm$ 59.51	88.15 $\pm$ 57.21	93.22 $\pm$ 45.54	156.38 $\pm$ 89.90	160.75 $\pm$ 86.71
	123	55.54 $\pm$ 29.93	84.77 $\pm$ 51.15	85.67 $\pm$ 50.33	97.57 $\pm$ 49.59	167.84 $\pm$ 103.11	170.45 $\pm$ 101.23
	456	52.95 $\pm$ 29.73	80.22 $\pm$ 52.41	81.97 $\pm$ 50.98	96.51 $\pm$ 46.44	165.49 $\pm$ 99.19	170.64 $\pm$ 96.10
	789	55.11 $\pm$ 31.02	84.76 $\pm$ 54.17	85.82 $\pm$ 53.24	98.40 $\pm$ 46.65	165.96 $\pm$ 97.21	170.64 $\pm$ 93.95
	1024	58.54 $\pm$ 32.09	93.05 $\pm$ 54.67	93.68 $\pm$ 53.94	97.75 $\pm$ 45.25	167.29 $\pm$ 87.88	169.68 $\pm$ 86.63

Diffusion Policy (iDiff-IL) [12], a joint image trajectory diffusion method using a dual-head UNet to denoise both spaces simultaneously; (5) Conditional Diffusion Policy (CondDiff) [12], a variant applying DDPM denoising in trajectory space only, conditioned on the image; and (6) Motion Indeterminacy Diffusion (MID) [7], a latent-space diffusion method that encodes frames into a context vector and applies a Transformer-based DDPM denoiser for trajectory generation; (7) VanillaCNN, as a strawman baseline implementing only a simple 4-layer CNN that directly regresses trajectory points from stacked input frames.

4.2 Result Analysis

Quantitative Results. Table 1 reports results under two experimental settings. SutureFormer consistently outperforms all baselines across all metrics and 5 random seeds. With Obs=6, Pred=3 (Obs = 6 denotes an observed surgical trajectory consisting of 6 annotated keyframes, with the remaining labels generated by interpolation at a frame interval of 1, whereas Pred = 3 denotes the prediction horizon containing 3 future keyframes; another setting is also the same), SutureFormer achieves an ADE of 52.95 (58.6% reduction over BC at 128.15) and FD of 80.22 (vs. 146.24 for BC), indicating substantially better trajectory shape fidelity. Under the more challenging Obs=3, Pred=6 setting, all methods degrade as historical observational information decreases, yet ours remains the best, achieving ADE 93.22, FDE 156.38 and FD 160.75. SutureFormer maintains clear advantages in ADE (31.3% reduction over BC), FDE (14.2% reduction) and FD (17.1% reduction), demonstrating that the CQL-based policy produces

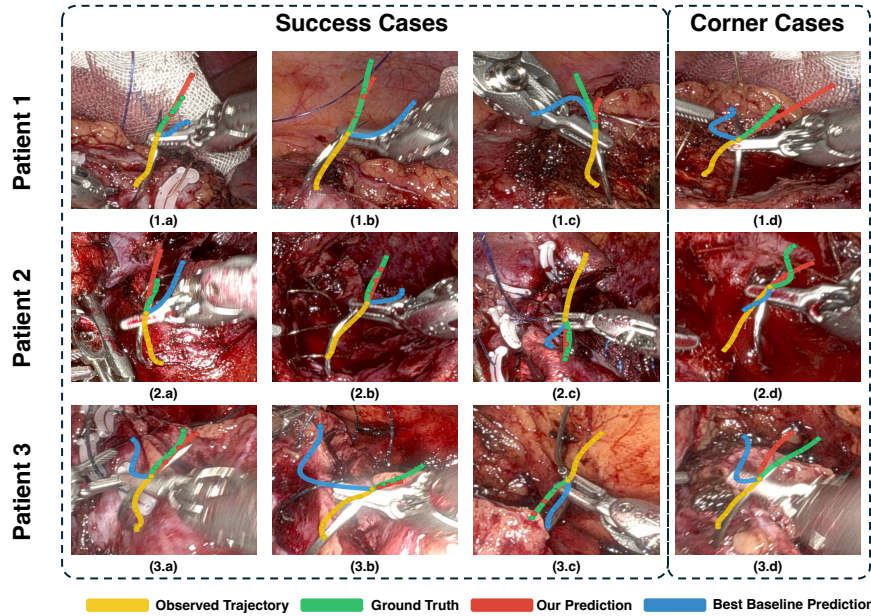


Fig. 2. Qualitative comparison of predicted trajectories on the testset. The **yellow** curve denotes the observed trajectory, the **green** curve represents the ground truth future trajectory, the **red** curve shows the prediction from our SutureFormer and the **blue** curve indicates the best baseline prediction.

more accurate trajectories with better shape consistency even under sparse observations. The goal-conditioned navigation formulation proves particularly beneficial when visual context is limited, as explicit target guidance compensates for reduced observational evidence. Beyond achieving superior performance, our method also exhibits lower per-trajectory standard deviation across all metrics and random seeds compared to all baselines, demonstrating stronger robustness and consistency in handling diverse surgical trajectory patterns. *Notably, the performance of these baselines differs substantially from prior studies, primarily for two reasons: first, we compute our evaluation metrics at the original image resolution, rather than downsampled one; and second, the surgical scenes in our kidney wound suturing dataset present considerably higher complexity.*

Qualitative Results. Fig. 2 illustrates representative predicted trajectories overlaid on surgical images, alongside the corresponding ground-truth paths and prediction results from state-of-art baseline method. As observed across diverse suturing scenarios from multiple patients, the baseline method frequently struggles to capture the complex spatial dynamics of the surgical instruments, resulting in predicted trajectories that significantly deviate from the true paths. In contrast, our proposed approach consistently maintains high fidelity to the ground truth. The proposed model generates smooth trajectories that conform to the general curvature of the suturing path.

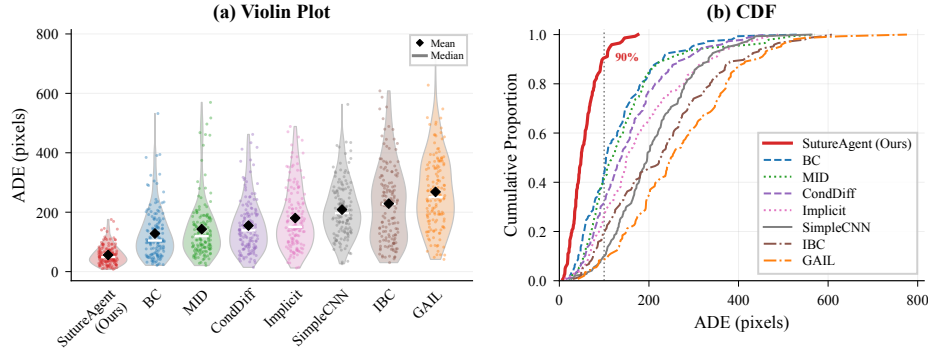


Fig. 3. Distribution of Average Displacement Error (ADE) across all methods on the test set. (a) Violin plot showing the ADE distribution for each method, with individual data points overlaid. Black diamonds indicate the mean and white horizontal lines indicate the median. (b) Empirical cumulative distribution function (CDF) of ADE. The dashed vertical line marks the ADE=100 pixel threshold, where our method achieves 90% of trajectories below this value.

Statistical Analysis. Fig. 3 presents the evaluated ADE distribution across all methods on the test set. SutureFormer achieves the lowest mean ADE of 55.99 pixels (median: 49.10), substantially outperforming the second-best method BC (mean: 128.15, median: 104.61) by a margin of 56.3%. As shown in the CDF plot, 90% of our predictions fall below an ADE of 100 pixels, compared to only 45% for BC. Wilcoxon signed-rank tests confirm that our method significantly outperforms all baseline methods ($p < 0.001$). The compact distribution in the violin plot further demonstrates that SutureFormer maintains consistently low prediction errors across diverse surgical trajectories, with markedly lower variance (std: 32.71) compared to other methods (std: 85.30–130.58).

Generalization Analysis. To evaluate the generalisability of the learned Q-function across trajectories of varying lengths, we visualise the per-step Q-value curves on four randomly selected test trajectories with prediction horizons $T \in \{\text{short, medium-short, medium-long, long}\}$, as shown in Fig. 4. Across all horizons, $Q_{\text{policy}}(s_k, a_k^\pi)$ consistently meets or exceeds $Q_{\text{expert}}(s_k, a_k^*)$, indicating that the CQL conservative penalty has successfully shaped the value function to prefer the policy’s actions over suboptimal offline demonstrations. Both curves exhibit a monotonically decreasing trend as k increases, which is consistent with the discounted return structure of the reward function. Notably, this behaviour is preserved even for the longest trajectory ($T = 48$), demonstrating that the Q-network generalises stably across variable-length surgical sequences without degradation in value estimation quality.

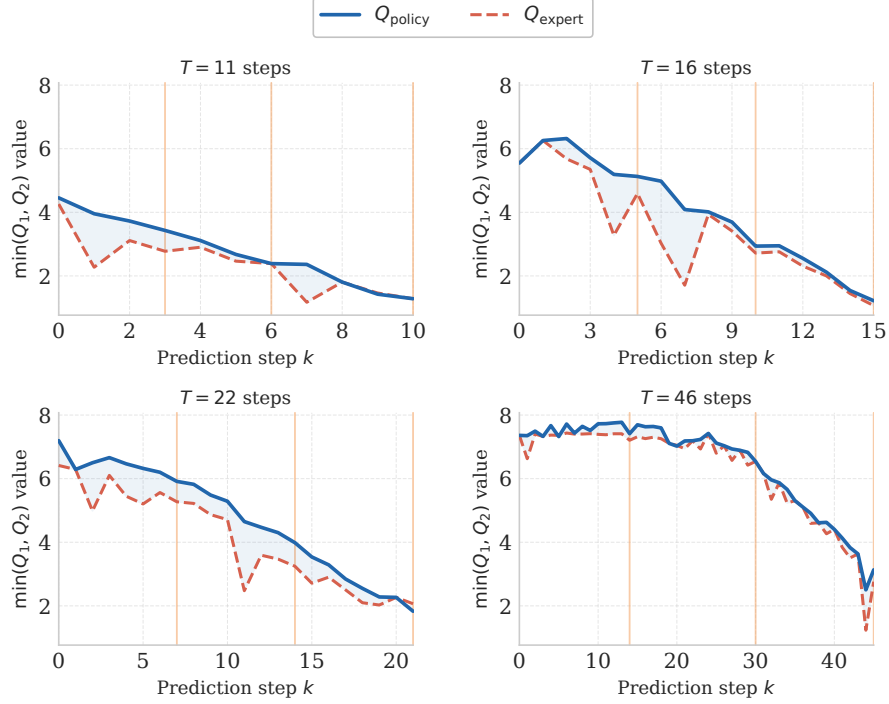


Fig. 4. Per-trajectory Q-value curves on four test trajectories of increasing prediction horizon, demonstrating generalisation across variable-length sequences. $Q_{\text{policy}}(s_k, a_k^{\pi})$ (solid blue) is the pessimistic value estimate $\min(Q_1, Q_2)$ of the policy’s chosen action at step k ; $Q_{\text{expert}}(s_k, a_k^*)$ (dashed red) is the value of the corresponding ground-truth expert action. Orange vertical lines indicate keyframe positions. $Q_{\text{policy}} \geq Q_{\text{expert}}$ consistently across all horizons confirms that the CQL conservative penalty successfully shapes the Q-function to rank the learned policy above suboptimal offline actions, regardless of trajectory length.

5 Conclusion

This paper introduces SutureFormer, a novel framework that reformulates surgical trajectory prediction in pixel space based on goal-conditioned offline reinforcement learning using Conservative Q-Learning. By requiring only 9 keyframe annotations per trajectory and operating without robot kinematics, the framework demonstrates broad applicability to existing clinical video archives. Experimental results on 1,158 trajectories demonstrate that SutureFormer significantly outperforms best diffusion and imitation learning baselines, achieving up to a 58.6% reduction in ADE. Future work will extend SutureFormer to broader laparoscopic procedures and validate its performance through ex-vivo porcine experiments on robotic platforms, facilitating its transition toward precise, advancing its translation toward real-world cognitive surgical assistance.

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